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CS471 Project 3

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1 Introduction

Benchmark objective functions are widely used to evaluate numerical behavior in continuous optimization. Each function exhibits different landscape properties (e.g., unimodal vs. multimodal, separable vs. non-separable, smooth vs. rugged), which affects both (1) the distribution of objective values produced by random samples and (2) the computational effort required to evaluate fitness.
This project implements ten standard benchmark functions and evaluates them using randomized populations. Rather than running an optimizer, the goal is to measure *fitness evaluation outcomes*: objective value statistics and evaluation time in milliseconds for computing fitness for an entire population. This establishes a consistent baseline for comparing benchmark problems under identical sampling conditions.

2 System Design and Implementation

The program was refactored into modular components:

- **Population:** An $n \times m$ matrix ($\mathbb{R}^{n \times m}$), where n is the number of experiments (default $n = 30$) and $m \in \{10, 20, 30\}$ is the problem dimension. Each row corresponds to one candidate solution vector.
- **Fitness:** A vector in $\mathbb{R}^n$ storing the scalar objective function values returned from the problem evaluation.
- **Problem:** A collection of benchmark functions (Schwefel through Egg Holder). Input is a vector from the population; output is a scalar fitness value.

Configuration and I/O: All key settings are read from an external input file (dimension m, population size n, problem type 1–10, seed, output CSV path). Fitness values are written to a CSV file for external analytics.
Timing Measurement: The program measures wall-clock time (milliseconds) for evaluating the fitness of the *entire population* (i.e., timing includes only the loop that computes the fitness vector, excluding file I/O and configuration parsing). This timing value is stored in the output and later analyzed in Python.

3 Optimization Algorithms

This project evaluates two population-based metaheuristic algorithms: **Differential Evolution (DE)** and **Particle Swarm Optimization (PSO)**. Both algorithms were implemented in C and applied to the same benchmark functions, problem dimensions, and bounds to enable direct comparison of convergence behavior, solution quality, and computational cost.
Differential Evolution is a stochastic population-based optimization algorithm that generates new candidate solutions through mutation and crossover. Given a population of size N, each individual vector is mutated using scaled differences of other randomly selected population members.
The general DE mutation form is:

$$v = x_{r_1} + F(x_{r_2} - x_{r_3})$$

where F is the differential weight controlling step size. Crossover is then applied using crossover probability CR to generate a trial vector. If the trial vector yields improved fitness, it replaces the target vector (selection step).
In this project, the following ten DE strategies were implemented:

1. **DE/best/1/exp**
$$v = x_{best}^{(G)} + F \cdot \left(x_{r_2}^{(G)} - x_{r_3}^{(G)}\right)$$
2. **DE/rand/1/exp**
$$v = x_{r_1}^{(G)} + F \cdot \left(x_{r_2}^{(G)} - x_{r_3}^{(G)}\right)$$
3. **DE/rand-to-best/1/exp**
$$v = x_i^{(G)} + \lambda \cdot \left(x_{best}^{(G)} - x_i^{(G)}\right) + F \cdot \left(x_{r_1}^{(G)} - x_{r_2}^{(G)}\right)$$
4. **DE/best/2/exp**
$$v = x_{best}^{(G)} + F \cdot \left(x_{r_1}^{(G)} + x_{r_2}^{(G)} - x_{r_3}^{(G)} - x_{r_4}^{(G)}\right)$$
5. **DE/rand/2/exp**
$$v = x_{r_1}^{(G)} + F \cdot \left(x_{r_2}^{(G)} + x_{r_3}^{(G)} - x_{r_4}^{(G)} - x_{r_5}^{(G)}\right)$$
6. **DE/best/1/bin**
$$v = x_{best}^{(G)} + F \cdot \left(x_{r_2}^{(G)} - x_{r_3}^{(G)}\right)$$

7. **DE/rand/1/bin**
$$v = x_{r_1}^{(G)} + F \cdot \left(x_{r_2}^{(G)} - x_{r_3}^{(G)}\right)$$
8. **DE/rand-to-best/1/bin**
$$v = x_i^{(G)} + \lambda \cdot \left(x_{best}^{(G)} - x_i^{(G)}\right) + F \cdot \left(x_{r_1}^{(G)} - x_{r_2}^{(G)}\right)$$
9. **DE/best/2/bin**
$$v = x_{best}^{(G)} + F \cdot \left(x_{r_1}^{(G)} + x_{r_2}^{(G)} - x_{r_3}^{(G)} - x_{r_4}^{(G)}\right)$$
10. **DE/rand/2/bin**
$$v = x_{r_5}^{(G)} + F \cdot \left(x_{r_1}^{(G)} + x_{r_2}^{(G)} - x_{r_3}^{(G)} - x_{r_4}^{(G)}\right)$$

Where:

- F is the differential weight,
- λ is the scaling factor toward the best solution,
- $x_{best}^{(G)}$ is the best individual in generation G,
- r_1, r_2, r_3, r_4, r_5 are distinct random indices,
- `exp` and `bin` indicate exponential and binomial crossover.

These strategies vary in exploration and exploitation behavior. Strategies incorporating x_{best} emphasize exploitation, while purely random-based strategies promote exploration and population diversity.

3.1 Particle Swarm Optimization

Particle Swarm Optimization models collective behavior in a swarm. Each particle maintains:

- A position vector x_i
- A velocity vector v_i
- A personal best position $pbest_i$
- A global best position $gbest$

Velocity is updated according to:

$$v_i = wv_i + c_1 r_1 (pbest_i - x_i) + c_2 r_2 (gbest - x_i)$$

Position is then updated by:

$$x_i = x_i + v_i$$

where:

- w is inertia weight,
- c_1 is cognitive acceleration,
- c_2 is social acceleration,
- $r_1, r_2 \sim U(0,1)$.

PSO balances exploration and exploitation through the interaction between personal and global best information. Velocity clamping and boundary constraints were applied to ensure stability.

4 Benchmark Problems (Introductions)

The ten benchmark functions were selected to represent a variety of search landscapes:

1. **Schwefel (1):** Highly multimodal and deceptive; many local minima. Random samples often produce wide objective value ranges due to strong nonlinearities.
2. **De Jong 1 / Sphere (2):** Smooth, convex, unimodal baseline. Fitness values tend to cluster more tightly relative to multimodal functions.
3. **Rosenbrock (3):** Features a narrow curved valley; evaluation is simple but the landscape is nonconvex and nonseparable.
4. **Rastrigin (4):** Strong periodic structure with many local minima. Objective values often show high variance due to oscillatory cosine terms.
5. **Griewank (5):** Many regularly distributed local minima; combines a sum and a product term.
6. **Sine Envelope Sine Wave (6):** Oscillatory behavior with envelope scaling; typically produces rugged response surfaces.
7. **Stretch V Sine Wave (7):** Emphasizes nonlinear scaling and oscillations; sensitive to pairwise variable interactions.
8. **Ackley One (8):** Mixes exponential and trigonometric components; designed to be multimodal with broad flat regions.
9. **Ackley Two (9):** A more interaction-heavy variant; frequently increases computational cost due to additional nonlinear terms.
10. **Egg Holder (10):** Very rugged with steep ridges/valleys; often yields extreme values depending on input range and nonlinear structure.

Table 4.1: Experiments

	Name	$f(x^*)$	Dimension	Range
f_1	Schwefel	0	30	$[-512, 512]^n$
f_2	De Jong 1	0	30	$[-100, 100]^n$
f_3	Rosenbrock's Saddle	0	30	$[-100, 100]^n$
f_4	Rastrigin	0	30	$[-30, 30]^n$
f_5	Griewank	0	30	$[-500, 500]^n$
f_6	Sine Envelope Sine Wave	$-1.4915(n-1)$	30	$[-30, 30]^n$
f_7	Stretch V Sine Wave	0	30	$[-30, 30]^n$
f_8	Ackley One	$-7.54276-2.91867(n-3)$	30	$[-32, 32]^n$
f_9	Ackley Two	0	30	$[-32, 32]^n$
f_{10}	Egg Holder	–	30	$[-500, 500]^n$

5 Experimental Procedure

For each experiment:

1. Read configuration parameters:
   - Dimension $m \in \{10, 20, 30\}$
   - Population size
   - Algorithm type (DE strategy or PSO)
   - Number of generations
   - Parameter values (F, CR, λ, w, c_1, c_2)
   - Problem type
   - Bounds
   - Random seed
2. Initialize the population uniformly within problem bounds.
3. Run the selected algorithm for a fixed number of generations.
4. Record:
   - Best fitness per generation
   - Final best fitness
   - Total execution time (milliseconds)
5. Export results to CSV for statistical comparison.

Each configuration was executed multiple times with different seeds to evaluate stability and consistency.

Problem	DE/best/1/exp			DE/rand/1/exp			DE/rand-to-best/1/exp			DE/best/2/exp			DE/rand/2/exp		
	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time
Schwefel	**96.135**	103.666	12.481	169.665	40.807	12.011	574.674	78.563	12.971	318.668	103.303	12.915	390.585	60.298	12.422
DeJong1	**1.955**	0.604	2.135	22.527	7.455	2.138	1.989	0.556	2.735	74.671	21.773	2.408	139.754	40.557	2.434
Rosenbrock	1862.456	631.061	2.367	21339.464	9103.351	2.221	**1631.964**	701.064	2.464	137581.176	76261.069	2.423	493656.714	317303.481	2.454
Rastrigin	**19.098**	3.328	9.972	36.336	3.323	9.843	24.332	4.426	9.939	57.582	7.690	10.504	72.794	9.129	10.042
Griewangk	**0.746**	0.113	11.032	1.136	0.046	11.019	0.768	0.078	11.196	1.459	0.124	11.005	1.898	0.313	11.096
SineEnvSine	**-12.573**	0.164	9.503	-12.298	0.176	9.414	-12.454	0.153	9.468	-12.063	0.169	9.795	-11.922	0.243	9.970
StretchVSine	**10.524**	0.458	36.565	10.737	0.495	36.578	10.547	0.541	37.554	10.726	0.759	37.130	10.765	0.637	36.874
Ackley1	**-24.265**	1.017	15.627	-17.317	1.907	15.940	-21.462	1.473	15.472	-9.645	2.284	15.801	-5.655	3.400	15.653
Ackley2	**8.644**	1.866	33.541	24.163	2.873	33.495	9.350	1.514	33.892	41.976	3.668	33.939	54.034	6.280	31.664
EggHolder	**-6437.604**	340.166	17.637	-6012.164	188.380	17.702	-6032.478	240.188	17.970	-6005.230	226.880	18.022	-5767.845	167.151	18.018

Table 5.1: DE Results (10 Dimensions) – EXP

Problem	DE/best/1/exp			DE/rand/1/exp			DE/rand-to-best/1/exp			DE/best/2/exp			DE/rand/2/exp		
	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time
Schwefel	**1458.040**	144.106	21.927	1474.499	100.608	22.829	1957.032	157.326	23.146	1637.009	119.770	22.161	1696.849	116.225	21.804
DeJong1	746.546	133.453	2.805	1567.136	220.478	2.825	**636.120**	100.856	3.466	3211.377	528.808	3.077	4207.695	441.673	3.341
Rosenbrock	8224390.454	2986590.052	2.958	32874598.054	14247250.010	2.985	**6452566.925**	2657406.598	3.206	140129004.088	49810439.953	3.230	229805139.523	66602865.291	3.450
Rastrigin	221.705	16.848	18.621	309.601	23.352	18.440	**209.008**	21.424	18.387	480.139	49.390	18.329	555.897	74.135	18.747
Griewangk	5.587	0.894	20.275	11.007	1.318	20.352	**5.069**	0.709	21.045	20.151	4.443	20.475	25.450	4.841	20.985
SineEnvSine	**-23.421**	0.334	18.169	-22.885	0.363	18.639	-23.410	0.326	18.239	-21.998	0.384	18.517	-21.970	0.378	18.924
StretchVSine	35.104	2.407	75.894	35.601	3.270	76.559	**34.179**	2.900	76.800	37.998	4.249	76.520	38.808	4.685	76.177
Ackley1	14.015	5.346	31.156	32.824	6.723	31.586	**10.829**	6.196	31.177	58.313	7.700	31.695	69.988	11.722	31.872
Ackley2	**126.083**	7.022	65.895	170.937	9.260	65.897	129.728	7.839	66.997	222.887	10.957	64.794	230.236	12.756	64.370
EggHolder	**-10222.986**	499.593	35.396	-9888.256	409.378	35.294	-10044.506	453.175	35.901	-9845.527	457.581	35.549	-9813.532	356.179	36.022

Table 5.2: DE Results (20 Dimensions) – EXP

	DE/best/1/exp			DE/rand/1/exp			DE/rand-to-best/1/exp			DE/best/2/exp			DE/rand/2/exp		
Problem	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time
Schwefel	3357.229	281.522	34.361	**3183.908**	135.908	31.269	3961.602	337.914	31.504	3241.833	178.225	31.536	3222.062	164.448	33.478
DeJong1	5344.998	727.827	3.589	8737.537	1084.278	3.593	**5033.514**	671.023	4.190	13512.028	1958.096	3.927	15649.282	1936.130	4.305
Rosenbrock	331486521.110	118161107.229	3.668	686679201.758	150950768.994	3.698	**240914737.297**	75548977.550	4.048	1661531714.754	413180557.646	4.180	2225689310.818	694264903.196	4.594
Rastrigin	776.254	83.296	26.512	1041.090	104.954	26.073	**731.095**	64.684	26.179	1468.786	189.197	26.718	1670.701	170.153	26.846
Griewangk	34.459	3.825	29.574	53.493	7.018	29.699	**31.625**	3.924	30.223	88.655	8.866	30.780	99.979	12.868	30.344
SineEnvSine	-32.792	0.475	26.805	-32.365	0.570	26.841	**-33.049**	0.452	27.099	-30.858	0.602	27.284	-30.621	0.474	28.414
StretchVSine	75.576	6.575	113.302	78.950	6.193	114.291	**71.810**	5.951	116.782	84.135	8.725	113.282	86.596	9.537	114.412
Ackley1	105.792	10.102	46.204	131.789	14.914	48.238	**97.767**	10.186	47.278	185.860	11.008	46.504	190.378	16.078	47.031
Ackley2	325.353	9.785	99.697	360.838	11.273	98.419	**323.783**	10.338	99.333	412.038	13.987	96.983	419.079	13.038	94.826
EggHolder	**-13069.837**	610.103	53.104	-12995.474	505.325	52.341	-12810.096	522.289	53.399	-13011.490	511.955	53.405	-12889.429	487.245	54.001

Table 5.3: DE Results (30 Dimensions) – EXP

	DE/best/1/bin			DE/rand/1/bin			DE/rand-to-best/1/bin			DE/best/2/bin			DE/rand/2/bin		
Problem	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time
Schwefel	593.143	155.126	14.736	**459.145**	69.399	13.949	820.267	85.550	15.071	626.003	66.355	14.664	699.343	48.131	16.983
DeJong1	4.668	1.580	3.820	265.850	56.077	3.911	**3.931**	1.272	4.837	1288.211	354.524	4.700	2225.126	591.709	5.023
Rosenbrock	8435.534	4269.912	4.635	1179685.475	554292.440	3.872	**5773.303**	3314.401	4.044	25752168.627	12235908.329	4.434	90493864.464	58694479.632	4.402
Rastrigin	**42.693**	6.537	11.347	94.222	9.889	11.324	43.951	4.885	11.379	207.153	40.933	11.512	316.170	57.291	12.100
Griewangk	0.893	0.099	12.543	2.609	0.465	13.077	**0.854**	0.119	12.534	8.945	2.101	12.791	16.806	3.705	13.332
SineEnvSine	**-11.346**	0.312	10.976	-10.843	0.321	11.162	-11.311	0.215	11.153	-10.379	0.312	11.670	-10.074	0.284	11.663
StretchVSine	16.476	2.848	38.837	16.111	2.737	38.340	**15.629**	2.627	39.426	16.738	2.937	38.988	16.691	2.992	38.707
Ackley1	**-14.076**	2.374	16.980	3.331	3.816	17.328	-12.773	3.015	17.263	21.643	6.393	17.359	34.938	6.320	17.780
Ackley2	**17.719**	3.428	35.747	64.790	6.825	33.487	19.663	2.493	34.809	98.225	9.026	33.099	108.365	6.809	32.917
EggHolder	**-6316.199**	614.960	19.214	-5339.827	238.949	19.460	-5627.926	387.269	19.363	-5387.742	390.135	19.504	-5141.588	301.596	19.631

Table 5.4: DE Results (10 Dimensions) – BIN

	DE/best/1/bin			DE/rand/1/bin			DE/rand-to-best/1/bin			DE/best/2/bin			DE/rand/2/bin		
Problem	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time
Schwefel	**1672.155**	221.945	26.305	1961.099	121.736	27.489	2596.520	296.654	27.033	2006.271	131.303	27.936	2335.073	136.118	29.469
DeJong1	3554.213	688.267	6.648	12786.193	1757.066	6.898	**2755.611**	400.197	8.082	23697.650	3176.261	7.988	27646.703	3181.454	7.795
Rosenbrock	337026365.205	146289349.808	8.023	2350538641.771	429480085.339	7.351	**222108903.438**	79557685.966	6.930	6670772990.166	1951056845.587	7.638	8266193556.615	2233482677.736	8.046
Rastrigin	524.853	57.144	21.709	1272.795	227.533	21.956	**431.313**	52.787	21.874	2307.799	415.498	22.673	-	-	-
Griewangk	22.553	3.744	25.308	77.586	13.744	24.702	**18.799**	4.081	24.410	144.612	21.591	24.888	176.092	24.735	25.243
SineEnvSine	19.828	0.416	21.857	-18.845	0.441	22.008	**-19.877**	0.477	22.393	-17.987	0.451	22.556	-17.806	0.468	22.766
StretchVSine	88.467	11.107	80.530	87.926	11.676	80.772	**82.906**	11.463	80.898	107.263	8.691	81.153	101.708	15.590	80.819
Ackley1	70.830	11.898	35.329	141.621	16.574	35.552	**66.546**	9.285	35.061	201.674	14.168	36.150	218.236	19.124	36.194
Ackley2	**222.872**	11.232	69.113	280.756	7.733	68.644	232.608	11.609	68.860	299.218	7.968	67.989	305.521	6.310	66.864
EggHolder	**-8559.004**	1188.434	39.848	-7675.226	553.236	40.300	-7898.059	679.550	39.714	-7888.395	687.846	39.484	-7366.789	375.942	40.258

Table 5.5: DE Results (20 Dimensions) – BIN

	DE/best/1/bin			DE/rand/1/bin			DE/rand-to-best/1/bin			DE/best/2/bin			DE/rand/2/bin		
Problem	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time
Schwefel	**2938.847**	312.353	37.592	4052.754	266.331	39.328	4986.463	515.756	38.847	3872.101	353.216	39.185	4537.087	239.001	nan
DeJong1	21216.943	2561.281	9.435	42552.053	3528.628	9.848	**16566.443**	2274.402	10.872	53985.940	5996.041	11.861	55434.293	5757.416	11.187
Rosenbrock	5057089726.344	1576824023.563	9.711	16142728703.817	2992637893.066	9.984	**3878335031.965**	989802554.004	10.153	21145251844.137	3767578345.224	10.684	22589936570.440	3892913098.369	12.292
Rastrigin	2063.518	326.083	31.989	3967.249	337.283	32.076	**1950.860**	204.432	32.426	5402.910	365.380	33.653	-	-	-
Griewangk	124.553	18.910	38.696	265.231	16.697	36.773	**111.632**	12.940	36.326	359.189	28.969	36.531	359.528	33.174	36.835
SineEnvSine	-27.499	0.469	33.169	-26.390	0.626	32.843	**-27.681**	0.820	33.235	-25.450	0.375	34.061	-25.413	0.726	34.151
StretchVSine	**189.865**	16.551	120.781	207.567	15.394	120.547	190.983	16.029	122.170	221.326	17.725	119.952	216.839	19.549	121.105
Ackley1	230.347	20.285	51.919	337.962	15.753	53.470	**213.575**	20.973	53.337	403.905	17.799	53.573	417.049	19.821	53.778
Ackley2	**438.584**	16.367	104.127	476.256	8.902	102.893	442.015	12.167	103.373	498.550	8.618	99.603	499.752	10.509	99.678
EggHolder	**-9799.606**	1270.872	59.929	-8912.466	525.943	60.706	-9339.944	1039.764	59.975	-9215.929	777.327	59.408	-8916.303	596.669	59.697

Table 5.6: DE Results (30 Dimensions) – BIN

Table 5.7: PSO Results

Problem	**PSO (10 dim)**			**PSO (20 dim)**			**PSO (30 dim)**		
	Avg	Std	Time	Avg	Std	Time	Avg	Std	Time
Schwefel	627.840	209.228	11.201	2085.623	406.719	21.844	3878.388	818.043	32.536
DeJong1	10.154	11.988	2.543	242.892	167.329	4.960	884.871	373.581	7.435
Rosenbrock	42489.204	181236.172	2.630	940748.500	1362219.815	5.111	10337590.569	11727762.457	7.500
Rastrigin	44.502	10.601	10.283	177.010	21.527	20.351	369.199	50.943	29.992
Griewangk	0.965	0.157	11.291	2.149	0.897	22.538	5.949	2.206	34.007
SineEnvSine	-11.924	0.338	9.615	-22.366	0.484	20.097	-32.281	0.925	30.191
StretchVSine	11.139	1.357	38.102	43.359	8.437	81.618	88.483	14.575	121.726
Ackley1	-16.610	3.526	16.168	4.421	13.453	34.188	49.876	19.451	50.488
Ackley2	24.642	12.236	34.737	103.761	24.334	70.902	212.101	32.513	107.569
EggHolder	-5393.312	647.241	18.024	-9210.243	1221.082	37.834	-12455.794	1896.668	57.485

6 Analysis and Findings

6.1 Differential Evolution Strategy Comparison

The effectiveness of Differential Evolution depended strongly on mutation strategy and problem dimensionality.

For $m = 10$, `best/1/exp` frequently produced the strongest results. DE significantly outperformed PSO on Schwefel, De Jong 1, Rastrigin, Griewank, Stretch V Sine Wave, Ackley 1, Ackley 2, and Egg Holder ($p < 0.05$ in all cases). However, performance differences were not statistically significant for Rosenbrock and Sine Envelope Sine Wave.

At $m = 20$, strategy selection shifted toward `rand-to-best/1/exp` for several functions. In this dimension, PSO significantly outperformed DE on De Jong 1, Rosenbrock, Rastrigin, and Griewank, while DE retained advantages on Schwefel, Sine Envelope Sine Wave, Stretch V Sine Wave, Ackley 2, and Egg Holder. Ackley 1 showed no statistically significant difference ($p = 0.0524$).

At $m = 30$, performance divergence became more pronounced. PSO significantly outperformed DE on De Jong 1, Rosenbrock, Rastrigin, Griewank, and both Ackley functions. In contrast, DE maintained statistically significant advantages on Schwefel, Sine Envelope Sine Wave, Stretch V Sine Wave, and Egg Holder. All comparisons at this dimension were statistically significant ($p < 0.05$).

These results indicate that mutation strategy performance is highly dimension- and landscape-dependent rather than universally superior.

6.2 PSO Performance

PSO demonstrated competitive performance, particularly as dimensionality increased.

At lower dimension ($m = 10$), PSO was generally outperformed by DE, with statistically significant disadvantages on most benchmark functions. However, it remained competitive on Rosenbrock and Sine Envelope Sine Wave.

At $m = 20$, PSO began outperforming DE on several continuous unimodal and moderately multimodal functions, including De Jong 1, Rosenbrock, Rastrigin, and Griewank.

At $m = 30$, PSO showed clear superiority on multiple large-scale problems, achieving significantly lower fitness values than DE on De Jong 1, Rosenbrock, Rastrigin, Griewank, Ackley 1, and Ackley 2. However, PSO struggled comparatively on Schwefel, Stretch V Sine Wave, and Egg Holder.

These findings indicate that PSO scalability depends heavily on problem structure and that its effectiveness improves on certain high-dimensional continuous landscapes.

6.3 DE vs. PSO

The comparative analysis reveals that algorithm superiority depends on both dimension and objective landscape.

For lower dimensions ($m = 10$), DE was dominant across most benchmarks, with statistically significant advantages in eight of ten problems.

At intermediate dimension ($m = 20$), results were mixed, with performance advantages distributed across both algorithms.
At higher dimension ($m = 30$), PSO achieved superior performance on six of ten benchmarks, while DE retained advantages on four.
These results suggest that DE provides strong performance at lower problem sizes and on rugged landscapes, while PSO scales more effectively for certain high-dimensional continuous optimization tasks.

Table 6.1: Analysis for 10 dimensions

Problem	Strategy	DE	PSO	t-value	p-value	$p < 0.05$
Schwefel	best/1/exp	96.135	627.840	-11.384	3.22e-12	Yes
De Jong 1	best/1/exp	1.955	10.154	-3.809	6.70e-04	Yes
Rosenbrock	rand-to-best/1/exp	1631.964	42489.204	-1.237	2.26e-01	No
Rastrigin	best/1/exp	19.098	44.502	-12.610	2.69e-13	Yes
Griewank	best/1/exp	0.746	0.965	-6.174	9.89e-07	Yes
Sine Envelope Sine Wave	best/1/exp	-12.573	-12.498	-0.408	6.85e-01	No
Stretch V Sine Wave	best/1/exp	10.524	53.541	-15.023	5.60e-15	Yes
Ackley 1	best/1/exp	-24.265	21.396	-14.018	1.58e-14	Yes
Ackley 2	best/1/exp	8.644	36631736.635	-3.540	1.38e-03	Yes
Egg Holder	best/1/exp	-6437.604	-5478.715	-3.580	1.23e-03	Yes

Table 6.2: Analysis for 20 dimensions

Problem	Strategy	DE	PSO	t-value	p-value	$p < 0.05$
Schwefel	best/1/exp	1458.040	2085.623	-7.798	1.34e-08	Yes
De Jong 1	rand-to-best/1/exp	636.120	242.892	10.584	1.79e-11	Yes
Rosenbrock	rand-to-best/1/exp	6452566.925	940748.500	11.935	1.03e-12	Yes
Rastrigin	rand-to-best/1/exp	209.008	177.010	6.125	1.13e-06	Yes
Griewank	rand-to-best/1/exp	5.069	2.149	15.852	8.01e-16	Yes
Sine Envelope Sine Wave	best/1/exp	-23.421	-18.321	-5.994	1.51e-06	Yes
Stretch V Sine Wave	rand-to-best/1/exp	34.179	136.116	-15.147	4.31e-15	Yes
Ackley 1	rand-to-best/1/exp	10.829	-5478.715	2.021	5.24e-02	No
Ackley 2	best/1/exp	126.083	-9438.494	2.744	1.04e-02	Yes
Egg Holder	best/1/exp	-10222.986	-11966.288	4.483	6.60e-05	Yes

Table 6.3: Analysis for 30 dimensions

Problem	Strategy	DE	PSO	t-value	p-value	$p < 0.05$
Schwefel	best/1/bin	2938.847	3878.388	-6.079	1.28e-06	Yes
De Jong 1	rand-to-best/1/exp	5033.514	884.871	26.429	7.65e-22	Yes
Rosenbrock	rand-to-best/1/exp	240914737.297	10337590.500	16.386	3.37e-16	Yes
Rastrigin	rand-to-best/1/exp	731.095	369.200	23.325	2.45e-20	Yes
Griewank	rand-to-best/1/exp	31.625	5.949	34.205	5.42e-25	Yes
Sine Envelope Sine Wave	rand-to-best/1/exp	-33.049	-23.817	-7.146	1.70e-08	Yes
Stretch V Sine Wave	rand-to-best/1/exp	71.810	21673.734	-5.704	3.77e-06	Yes
Ackley 1	rand-to-best/1/exp	97.767	36631736.635	-3.312	2.61e-03	Yes
Ackley 2	rand-to-best/1/exp	323.783	635207155.042	-3.428	1.98e-03	Yes
Egg Holder	best/1/exp	-13069.837	-11966.288	-2.855	7.40e-03	Yes

7 Conclusion

This project implemented and evaluated Differential Evolution (with multiple mutation strategies) and Particle Swarm Optimization across ten continuous benchmark functions and three problem dimensions.

Results demonstrate that algorithm performance is highly dependent on both mutation strategy selection and dimensionality.

At lower dimensionality ($m = 10$), Differential Evolution outperformed PSO on the majority of benchmark functions. At moderate dimensionality ($m = 20$), performance differences were mixed. At higher dimensionality ($m = 30$), PSO achieved superior performance on several large-scale continuous problems, while DE maintained advantages on specific rugged landscapes.

These findings highlight that no single population-based optimization algorithm is universally dominant. Instead, performance depends on the interaction between search dynamics, dimensionality, and landscape structure.